Helix Forge Research Uncertainty Analysis Methods Study

HELIX FORGE LAB / p04-full-005813 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded uncertainty analysis evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000252 (notebooks/exports/2026-07-13/helix-forge-research/benchmark-design/specifications/helix-forge-research-benchmark-design-slices-v01.js

Slice	Records	Precision	Recall	F1
slice-01	217	0.894	0.862	0.878
slice-02	178	0.876	0.875	0.875
slice-03	177	0.893	0.851	0.871
slice-04	242	0.875	0.864	0.869
slice-05	198	0.892	0.877	0.884